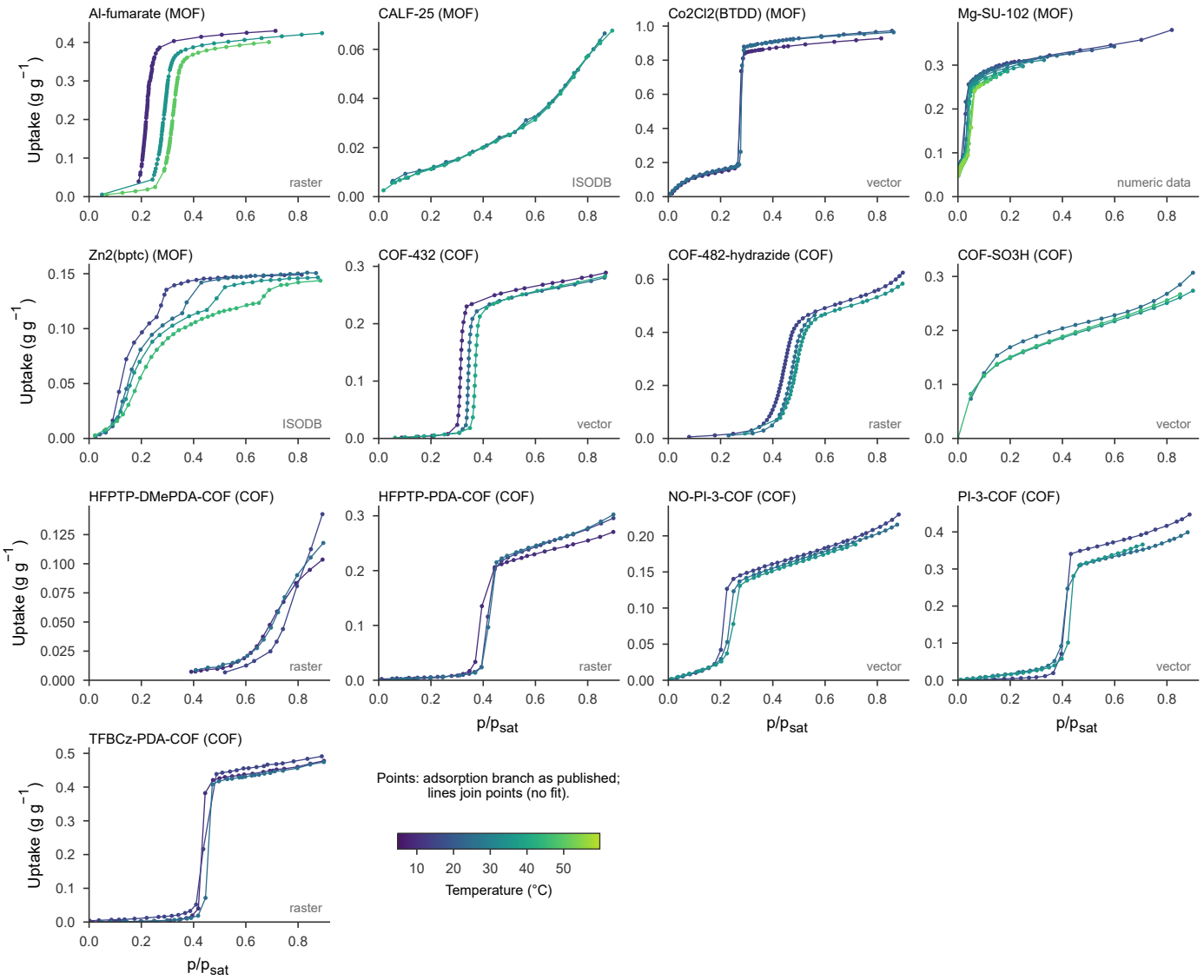
Supplementary Information for: Hard thermodynamic constraints and the temperature extrapolation of learned water isotherms in metal-organic and covalent organic frameworks: a pre-registered test

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Supplementary Table S1: Data

Material	Class	Set	Data	Source DOI	T range (K)	n_T	Points per T	p/p_{sat} range	q_{ref} (mol kg^{-1})
COF-432	COF	conf.	vector	10.1021/jacs.9b13094	283.15–313.15	3	28–29	0.063–0.870	16.04
COF-482-hydrazide	COF	conf.	raster	10.1021/acscentsci.2c00398	288.15–308.15	3	25–51	0.08–0.898	34.70
COF-SO_3H	COF	conf.	vector	10.1002/cssc.202301906	298.15–318.15	3	18–18	0.0003–0.900	17.04
HFPTP-DMePDA-COF	COF	conf.	raster	10.1038/s41467-021-27128-4	283.15–298.15	3	9–18	0.39–0.897	7.91
HFPTP-PDA-COF	COF	conf.	raster	10.1038/s41467-021-27128-4	283.15–298.15	3	24–32	0.0041–0.898	16.78
NO-PI-3-COF	COF	conf.	vector	10.1021/jacs.3c02572	288.15–308.15	3	30–37	0.0091–0.882	12.74
PI-3-COF	COF	conf.	vector	10.1021/jacs.3c02572	288.15–308.15	3	28–37	0.01–0.887	24.80
TFBCz-PDA-COF	COF	conf.	raster	10.1038/s41467-021-27128-4	283.15–298.15	3	29–30	0.0023–0.899	27.26
Al-fumarate	MOF	conf.	raster	10.1021/acscentsci.9b00745	283.15–323.15	3	56–59	0.049–0.892	23.86
CALF-25	MOF	conf.	ISODB	10.1021/ja306812r	298.00–313.00	3	17–20	0.018–0.894	3.75
Co_2Cl_2(BTDD)	MOF	conf.	vector	10.1021/acscentsci.7b00186	283.00–298.00	3	31–34	0.0072–0.862	53.97
Mg-SU-102	MOF	conf.	numeric	10.1038/s41467-025-59551-2	288.15–328.15	9	21–21	0.0013–0.819	21.11
Zn_2(bptc)	MOF	conf.	ISODB	10.1021/ja060946u	288.00–318.00	4	25–33	0.023–0.885	8.38
TFPPy-PDA-COF	COF	dev.	raster	10.1038/s41467-021-27128-4	283.15–298.15	3	7–15	0.46–0.900	35.41
CuBTC	MOF	dev.	ISODB	10.1016/j.ijheatmasstransfer.2012.07.068	293.00–358.00	9	23–41	0.0029–0.508	25.29
MIL-100	MOF	dev.	ISODB	10.1016/j.ijheatmasstransfer.2012.07.068	293.00–358.00	9	25–35	0.011–0.603	18.11
MIL-160	MOF	dev.	ISODB	10.1002/adma.201502418	293.00–313.00	3	32–54	0.0052–0.884	21.79

Supplementary Table S1. The 13 confirmatory materials (conf.) and the four development materials (dev.) after the inclusion rules and the preprocessing. Data: how the points were obtained (vector or raster digitisation of the published figure, numeric source data, or the NIST adsorption database, ISODB). n_T: number of temperatures; points per temperature with $p/p_{\mathrm{sat}} \leq 0.90$; q_{ref}: the largest uptake over all temperatures in this range, used only for scoring. Every decision and its reason is in data/inclusion_log.csv.



Supplementary Fig. S1. The confirmatory data: the adsorption isotherms of the 13 materials at every temperature, as used (relative pressure up to 0.90), coloured by temperature; points as published, joined by thin lines with no fit. The label in each panel gives how the points were obtained.

Supplementary Tables S2 to S7: Results in detail

			$R_m = E_P/E_X$						
Material	Class	n_{train}	H1b V-pen	H1c A-hard	H1d F-none	H1e C1	H1a F-hard	V-none	F-rel
COF-432	COF	2	0.86	0.83	0.37	0.98	0.70	0.84	1.58
COF-482-hydrazide	COF	2	0.95	1.18	0.35	1.03	0.22	0.96	0.72
COF-SO_3H	COF	2	10.91	9.92	6.19	8.58	6.04	9.80	5.35
HFPTP-DMePDA-COF	COF	2	1.23	4.60	0.51	0.74	1.14	0.94	0.95
HFPTP-PDA-COF	COF	2	0.99	2.28	1.01	1.01	0.85	1.00	0.95
NO-PI-3-COF	COF	2	2.47	1.55	1.07	2.47	0.20	2.09	2.69
PI-3-COF	COF	2	1.10	4.81	1.50	2.20	0.22	1.45	1.99
TFBCz-PDA-COF	COF	2	0.78	3.14	0.99	0.97	0.39	0.94	0.83
Al-fumarate	MOF	2	0.87	0.49	0.28	0.90	0.39	0.94	1.19
CALF-25	MOF	2	0.94	0.98	0.39	0.59	0.77	0.90	0.41
Co_2Cl_2(BTDD)	MOF	2	1.44	1.13	1.02	0.92	0.59	1.38	1.92
Mg-SU-102	MOF	8	1.01	0.73	0.24	1.56	0.28	1.07	0.31
Zn_2(bptc)	MOF	3	1.27	0.32	0.66	1.66	0.49	1.34	0.67
Hodges-Lehmann R			1.10	1.51	0.63	1.23	0.49	1.14	1.13

Supplementary Table S2. Error ratio of P against each comparator per material, $R_m = 10^{d_m}$, with d_m the difference of the mean of $\log_{10} E$ over E-high and E-low; below one, P extrapolated better. n_{train}: the smallest number of training temperatures over the two tasks. The last row is the Hodges-Lehmann estimate over the 13 materials (main text, Fig. 2a). H1a, V-none and F-rel are descriptive comparisons.

Model	Fits	In domain	Wilson 95 %	Band	Strict	All tasks	Class
P	26	0	0.00–0.13	0	0	0/39	none
F-hard	26	0	0.00–0.13	0	0	0/39	none
A-hard	26	0	0.00–0.13	0	0	0/39	none
C1	26	0	0.00–0.13	0	0	0/39	none
P-K	2	0	0.00–0.66	0	0	0/3	none
V-pen	26	9	0.19–0.54	10	11	12/39	some
V-none	26	8	0.17–0.50	9	10	12/39	some
F-pen	26	13	0.32–0.68	12	24	19/39	often
A-pen	26	16	0.43–0.78	16	21	25/39	often
A-none	26	15	0.39–0.74	15	19	24/39	often
F-rel	26	22	0.66–0.94	22	26	34/39	often
F-none	26	26	0.87–1.00	26	26	39/39	often

Supplementary Table S3. Violating fits per model over the 26 fits of E-high and E-low. In domain: the pre-registered count, a fall of more than 2 % of q_{ref} below the running maximum or an uptake below -2 % of q_{ref}, inside the penalty domain; Wilson 95 % interval of its rate. Band: the same rule applied to the rest of the grid, up to 20 K beyond the measured temperatures and to 0.95 p_{sat}. Strict: any decrease of uptake with pressure or any negative uptake anywhere on the grid, with no tolerance. All tasks: the in-domain count adding I-mid. Class: the pre-registered word.

Model	Fits	Median range	Largest range	Range above 1	Selected best	Selected worst
P	26	0.04	2.34	1	6	9
V-pen	26	0.11	1.46	1	5	5
V-none	26	0.13	1.76	2	10	7
A-hard	26	0.01	0.28	0	12	5
A-pen	26	0.03	0.54	0	7	4
A-none	26	0.04	1.41	2	6	4
C1	26	0.01	2.84	1	4	2
F-hard	26	0.54	1.48	5	7	7
F-pen	26	0.67	1.28	3	4	2
F-none	26	0.26	0.95	0	8	5
F-rel	26	0.18	0.97	0	3	10
P-K	2	0.34	0.47	0	0	1

Supplementary Table S4. Seed spread of the held-out error over the E-high and E-low fits. Range: the difference between the largest and the smallest $\log_{10} E$ over the 5 seeds of one fit, in decades; Range above 1: fits whose seeds differ by more than a factor of ten. Selected best (worst): fits in which the seed with the lowest training loss, the one reported, had the best (worst) held-out error of the five.

Truth	Comparator X	R	95 % interval	Holm p	Failed P/X	Verdict
KS	V-pen	0.827	0.74–0.978	0.024	0/0	differ, size uncertain
	A-hard	0.278	0.155–0.42	0.004	0/0	P better
	F-none	0.0259	0.0102–0.0785	0.004	0/5	P better
DA	V-pen	0.982	0.766–1.21	1.000	0/0	equivalent
	A-hard	114	72.4–181	0.004	0/0	A-hard better
	F-none	0.0561	0.0187–0.163	0.004	0/2	P better
FG	V-pen	1.27	0.266–1.7	1.000	0/3	inconclusive
	A-hard	0.41	0.363–0.545	0.004	0/0	P better
	F-none	0.0322	0.00624–0.122	0.004	0/6	P better
HS	V-pen	0.658	0.528–0.837	0.017	0/0	P better
	A-hard	0.22	0.0913–0.423	0.004	0/0	P better
	F-none	0.0269	0.00575–0.0843	0.004	0/5	P better
TC	V-pen	0.911	0.67–1.17	1.000	0/0	inconclusive
	A-hard	0.023	0.0124–0.0436	0.004	0/0	P better
	F-none	0.00687	0.00198–0.026	0.004	0/4	P better

Supplementary Table S5. The fifteen isosteric-heat tests on the synthetic truths (Supplementary Note 2). R: Hodges-Lehmann ratio of the root-mean-square error of Q_{st} at matched loading, P over X, over the 13 materials, with its exact 95 % interval; Holm p over the fifteen tests; Failed: materials on which a task failed (an undefined or non-finite heat, too few states in the loading window), ranked worst.

Dataset	n	H1b V-pen	H1c A-hard	H1d F-none	H1e C1
confirmatory set	13	1.10 [0.93, 1.65] inc	1.51 [0.84, 2.87] inc	0.63 [0.40, 1.22] inc	1.23 [0.92, 1.96] inc
coverage rule, first version	6	0.97 [0.83, 1.27] eq	0.88 [0.32, 1.48] inc	0.46 [0.32, 1.61] inc	1.11 [0.74, 3.20] inc
coverage rule, second version	9	1.19 [0.94, 3.32] inc	1.76 [0.73, 3.92] inc	1.00 [0.49, 2.48] inc	1.49 [0.95, 2.95] inc
no coverage rule	23	1.05 [0.97, 1.20] eq	1.52 [1.06, 2.19] inc	0.59 [0.42, 0.79] P	1.16 [0.97, 1.58] inc
7 points per isotherm	14	1.05 [0.89, 1.54] inc	1.29 [0.73, 2.67] inc	0.63 [0.41, 1.04] inc	1.20 [0.91, 1.66] inc
Co_2Cl_2(BTDD) from its characteristic curves	13	1.04 [0.93, 1.58] inc	1.51 [0.84, 2.87] inc	0.69 [0.40, 1.32] inc	1.28 [0.96, 2.03] inc
digitisation shift, copy 1	13	1.02 [0.90, 1.77] inc	1.35 [0.86, 3.06] inc	0.58 [0.38, 1.07] inc	1.35 [0.94, 2.22] inc
digitisation shift, copy 2	13	1.15 [0.96, 1.72] inc	1.34 [0.72, 2.29] inc	0.53 [0.37, 0.78] P	1.35 [0.97, 1.82] inc

Supplementary Table S6. The H1 comparisons on the pre-registered dataset variants: R with its exact 95 % interval and the verdict (P, P better; X, comparator better; eq, equivalent within 25 %; inc, inconclusive). The coverage variants are earlier versions of the coverage rule and the dataset without it; the digitisation shifts move every digitised isotherm by ± 0.002 in p/p_{sat}.

Comparison	X	R	Bootstrap 95 %	R, MOFs ($n = 5$)	R, COFs ($n = 8$)
H1b	V-pen	1.10	0.94–1.58	1.09	1.10
H1c	A-hard	1.51	0.87–2.67	0.69	2.67
H1d	F-none	0.63	0.42–1.07	0.43	0.93
H1e	C1	1.23	0.94–1.91	0.99	1.47
H1a (descriptive)	F-hard	0.49	0.33–0.82	0.48	0.50
H1b by source paper	V-pen	1.12	exact 0.92–3.08	10 sources	

Supplementary Table S7. The H1 comparisons with the percentile bootstrap interval (pre-registered as a secondary), split into MOFs and COFs (pre-registered subgroup, Hodges-Lehmann estimates only, no intervals at these sizes), and, as an exploratory check of the dependence between materials from one paper, H1b with the per-material differences averaged within each source paper (ledger X-4).

Supplementary Note 1: Can each verdict fail? Synthetic worlds and power

Before the freeze, the whole pipeline (the tuned configurations, fitting from 5 seeds, the violation counts and the analysis code) was run in synthetic worlds that share no uptake data with the confirmatory set. Each world has 13 materials; material i uses the temperatures of confirmatory material i. An expected outcome was written for each world before it ran. In KS0 the truth is a two-site Sips isotherm with van't Hoff affinities, inside the family of P; in DA it is a Dubinin-Astakhov curve, inside the family of the hard characteristic curve (A-hard); in FW it is a Kirchhoff-Sips isotherm with a large heat capacity, so that van't Hoff is wrong; FLAT has one weakly curved Langmuir site; SUB has a sub-linear Sips isotherm outside the family of P and inside that of the penalty model V-pen.

The comparison of P with A-hard discriminated in both directions: P extrapolated better in KS0 ($R = 0.52$, exact 95 % interval 0.35 to 0.70) and A-hard better in DA ($R = 1.26$, 1.07 to 1.49). Three expectations were missed. In FW the comparison of P with the free temperature network F-hard still read "P better" ($R = 0.33$): with two training temperatures the free network cannot extrapolate, so this comparison cannot fail and is reported descriptively. In FLAT the unconstrained network F-none violated physics "often". In SUB, built for V-pen to win, P was slightly better ($R = 0.96$, 0.94 to 0.99); the mixture of Langmuir terms in P approximates a sub-linear isotherm over the measured range. The accuracy comparison of hard constraints with penalties therefore could not be made to fail synthetically, and it is tested as non-inferiority.

The power of each verdict was simulated with the frozen analysis code at $n = 13$ (2000 simulations per setting). At the spread expected for the comparison with V-pen (0.158 decades), the non-inferiority criterion is met with probability 0.0225 when the true ratio is at the margin, $R = 1.25$, and 0.51 (normal) or 0.74 (Student t, 3 degrees of freedom) when the two models are equally accurate. A "not met" outcome is therefore likely even under equal accuracy and is never read as evidence that the constraints cost accuracy.

Supplementary Note 2: Synthetic truths for the isosteric heat

The heat test uses five truth families with exact heats. Their shape parameters are fitted by bounded least squares to each confirmatory material's middle-temperature isotherm; their thermodynamic parameters are fixed in advance: KS, a Langmuir site with $Q_0 = 55$ kJ mol^{-1} plus a cooperative Sips site with $Q_0 = 48$ kJ mol^{-1}, a heat capacity of 40 J mol^{-1} K^{-1} and an exponent of at least 1.5, inside the family of the Kirchhoff variant of P and near that of P; DA, a Dubinin-Astakhov curve $q = W_0 \exp[-(A/E)^n]$ with $1 \le n \le 6$, inside the family of the characteristic curve; FG, a Fowler-Guggenheim isotherm with $Q_0 = 45$ kJ mol^{-1} and a lateral attraction between 0 and $3.9\,RT_{\min}$, inside neither; HS, strong hydrophilic sites, a Langmuir site with $Q = 80$ kJ mol^{-1} holding at most 0.3 q_{ref} plus a Sips site with $Q = 46$ kJ mol^{-1}, inside the family of P with a heat that falls steeply with loading; TC, a Sips isotherm with $Q = 50$ kJ mol^{-1} whose capacity falls by 0.2 % per kelvin, inside the family of P and outside that of the characteristic curve. Gaussian noise of 0.4 % of q_{ref} is added with a fixed seed per material and family.

Supplementary Note 3: Digitisation error

The digitiser was tested against published figures whose numeric data are also published, before any model was fitted (logs/digitisation_error.json, logs/digitisation_validation.json). On the vector figure of Mg-SU-102[1] the extraction from the PDF drawing commands recovered 198 of 198 points with a root-mean-square uptake error of 7×10^{-6} g g^{-1} and a largest error of 5×10^{-5} g g^{-1}. The same figure rasterised at 150, 300 and 600 dpi and read by template

matching gave root-mean-square uptake errors of 6.3×10^{-4}, 4.0×10^{-4} and 3.5×10^{-4} $\mathrm{g\,g^{-1}}$, with no false detection; markers drawn hollow are not read by the raster method. On raster figures of HKUST-1 from another paper[2], the root-mean-square uptake error per series was 0.11 to 0.19 % of the axis span, or 0.0012 to 0.0034 of the series' largest uptake, again with no false detection; markers hidden under other markers are missed (recall of the visible markers 0.91 to 1.00). On vector figures of two further papers[3, 4] read through their legend colours, every point was recovered and the root-mean-square uptake error was 0.04 to 0.08 % of the axis span. The noise level of the synthetic tests, 0.4 % of q_{ref}, is the largest raster error per isotherm maximum rounded up.

Supplementary Note 4: Model definitions

All models map $s = \ln(p/\mathrm{Pa})$ and T to the uptake q ($\mathrm{mol\,kg^{-1}}$); the code is src/p3a/models.py, and the tuned settings and parameter counts are in Supplementary Table S8. $T_{\mathrm{ref}} = 300$ K throughout. The same symbol q is used for the uptake of every model; in the main text q^* marks the uptake of P written as a derivative.

P (PotentialIsotherm, van't Hoff). The reduced spreading pressure is

$$\psi(s,T) = \sum_k \frac{c_k(T)}{g_k}\,\mathrm{softplus}\big(g_k(s-\theta_k(T))\big), \qquad q = \frac{\partial\psi}{\partial s} = \sum_k c_k(T)\,\sigma\big(g_k(s-\theta_k(T))\big),$$

with σ the logistic function. Each term is therefore a Langmuir term ($g_k = 1$, the n_{H} Henry terms) or a cooperative Sips-type term ($g_k = 1.5 + \mathrm{softplus}(\cdot)$, the n_{c} cooperative terms) in $K_k(T)p$, with $\ln K_k = -\theta_k$. The capacities are $c_k(T) = c_{k0}\exp[-\alpha_k(T - T_{\mathrm{ref}})]$ with $c_{k0} > 0$ (a softplus), and the positions follow van't Hoff, $\theta_k(T) = \theta_{k0} - (Q_k/R)(1/T - 1/T_{\mathrm{ref}})$. The heats Q_k and the expansion coefficients α_k are unconstrained in sign. Positive capacities make q strictly increasing in s; the Henry terms give $q/p \to \sum_{\mathrm{Henry}} c_k K_k > 0$ as $p \to 0$; the heat $Q_{\mathrm{st}} = -RT^2(\partial q/\partial T)/(\partial q/\partial s)$ has a positive denominator and is finite everywhere, but its sign and size are not constrained. P is thus a positive mixture of Langmuir and cooperative terms with van't Hoff affinities, written as the derivative of one convex potential; C1 is a two-term member of the same broad family with free exponents.

V-none and V-pen. The same function, the same initial function for a given seed and the same tuned structure as P, but with signed capacities and every exponent free (including those of the Henry terms). V-pen adds the penalties below with weight λ.

P-K. As P, with a constant heat capacity Δc_k per term: $\theta_k(T) = \theta_{k0} - (Q_k/R)(1/T - 1/T_{\mathrm{ref}}) - (\Delta c_k/R)[\ln(T/T_{\mathrm{ref}}) + T_{\mathrm{ref}}/T - 1]$.

F-hard. The same potential and shape constraints as P, but $c_k(T)$ and $\theta_k(T)$ are outputs of a one-hidden-layer tanh network of the scaled inverse temperature $\hat{\tau} = (T_{\mathrm{ref}}/T - 1)/0.1$, with each θ_k kept within a bounded range around the data.

A-hard, A-none, A-pen. The characteristic curve $q = W(A)$, $A = RT\ln(p_{\mathrm{sat}}(T)/p)$, with $W(A) = \sum_k c_k\,\sigma(-g_k(A/A_{\mathrm{s}} - A_k))$, $A_{\mathrm{s}} = 2$ $\mathrm{kJ\,mol^{-1}}$. In A-hard, $c_k > 0$ and $g_k > 0$, so W decreases with A and q increases with p at every temperature; the model is temperature-invariant in A by construction, has no Henry limit, and its heat is exactly $\Delta_{\mathrm{vap}}H(T) + A$. A-none has signed c_k and free g_k; A-pen adds the monotonicity penalty.

F-none, F-pen, F-rel. A fully connected tanh network with inputs $((s - s_c)/s_w, \hat{\tau})$ and one linear output scaled by the largest training uptake (F-none, F-pen), or with $\ln(p/p_{sat})$ in place of s (F-rel); width and depth in Table S8.

C1. $q = \sum_{i=1}^{2} q_{m,i}\,\sigma\big(g_i(s + \ln b_i(T))\big)$, that is $q_{m,i}(b_i p)^{g_i}/[1 + (b_i p)^{g_i}]$, with $q_{m,i} > 0$, $g_i = 0.2 + \mathrm{softplus}(\cdot)$ and $\ln b_i(T) = \ln b_{i,\mathrm{ref}} + (Q_i/R)(1/T - 1/T_{\mathrm{ref}})$.

Penalties. On a collocation grid of 60 pressures (1 Pa to 0.9 p_{sat}) at 9 temperatures from $T_{min}-15$ K to $T_{max} + 15$ K, the monotonicity penalty is the mean of $\max(0, -\partial q/\partial s)^2$ divided by the squared uptake scale. The Henry penalty, at 0.01 Pa, is the mean of $[(\partial q/\partial s - q)/(|\partial q/\partial s| + |q|)]^2$ plus the same relative measure of negative uptake; it is zero only for an isotherm linear in p. The characteristic-curve models have no Henry penalty.

Fitting. The loss is the mean squared error of q divided by the largest training uptake, plus λ times the penalty and an optional ℓ_2 term; Adam (learning rate 10^{-2}) for the number of steps in Table S8, then L-BFGS for 300 iterations, in double precision from 5 seeds, keeping the seed with the lowest training loss. Inputs are centred on the training mean of s and scaled by its standard deviation (at least 0.5).

Model	Role	Form	Tuned settings	Parameters
P	confirmatory reference	potential, van't Hoff	n_{H} 2, n_{c} 2, Adam steps 2000	18
V-pen	H1b (primary)	as P, constraints as penalty	n_{H} 2, n_{c} 2, λ 0.001, Adam steps 2000	20
V-none	descriptive	as P, unconstrained	n_{H} 2, n_{c} 2, Adam steps 2000	20
A-hard	H1c	characteristic curve	n 6, Adam steps 4000	18
A-pen	H3 only	as A-hard, penalty	n 6, λ 0.001, Adam steps 4000	18
A-none	H3 only	as A-hard, unconstrained	n 6, Adam steps 4000	18
F-none	H1d	network in $(s, 1/T)$	width 32, depth 3, ℓ_2 0.0001, Adam steps 4000	2241
F-pen	H3 only	as F-none, penalty	width 32, depth 3, ℓ_2 0.0001, λ 0.1, Adam steps 4000	2241
F-hard	H1a (descriptive)	potential, network in $1/T$	n_{H} 1, n_{c} 2, width 8, Adam steps 2000	72
C1	H1e	dual-site Sips, van't Hoff	Adam steps 4000	8
F-rel	descriptive	network in $(\ln p/p_{\mathrm{sat}}, 1/T)$	width 32, depth 3, Adam steps 2000	2241
P-K	descriptive	P with heat capacity	n_{H} 2, n_{c} 2, Adam steps 2000	22

Supplementary Table S8. The models, their role in the analysis, the settings chosen on the development set (Methods), and the number of fitted parameters. n_{H}, n_{c}: Henry and cooperative terms; n: terms of the characteristic curve; width and depth of a network; ℓ_2: weight decay; λ: penalty weight. Every fit then ran L-BFGS for 300 iterations.

Supplementary References

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